

Field Maintenance Summary — FMS-2025-001

Document ID: FMS-2025-001

Reporting Period: 2025-01-01 to 2025-01-31

Site / Region: NW-PowerPool-WIND-024 (primary); secondary references WIND-014, WIND-003

Field Supervisor: Diego Mendoza-Ruiz, Lead Field Supervisor, NW-PowerPool

Distribution: Regional Maintenance Manager (Karen Hsu); Operations Planning; Procurement Liaison; on file at WIND-024 control building

Section 1 — Reporting Period Summary

January started ugly. The Q1 FY25 budget freeze landed on us on January 6 and we lost authority to issue any non-emergency WO above \$25K without sign-off from Portland. Practical effect for me: a stack of 14 preventive work orders that were on the January-February schedule are now deferred. The list is at the bottom of this section. I have flagged it to Karen twice already.

WO statistics, January 2025, NW-PowerPool-WIND-024:

Category	Planned	Completed	Deferred
Preventive WOs	19	5	14
Corrective WOs	7	7	0
Emergency WOs	0	1	n/a
Inspections	11	11	0

We completed all inspections — the freeze doesn't apply to inspection rounds, only to dispatched maintenance — but five of the eleven inspection reports flagged "PM recommended, deferred per freeze." That language is going to be a problem for us in any audit.

Deferred work orders (cited for the record):

- WO-2025-00118 — WIN-10131 gearbox oil and filter, scheduled 1/14
- WO-2025-00119 — WIN-10011 gearbox oil and filter, scheduled 1/16
- WO-2025-00124 — WIN-10131 vibration baseline, scheduled 1/21
- WO-2025-00127 — WIN-10058 lubrication service (cross-site, WIND-014)

- WO-2025-00131 — WIN-10016 yaw-bearing greasing
- WO-2025-00133 — SUB-10641 transformer oil sample
- WO-2025-00137 — TRA-10561 bushing inspection (cross-site, WIND-003)
- WO-2025-00142 — WIN-10011 pitch system actuator check
- WO-2025-00146 — WIN-10131 cooling-loop coolant exchange
- WO-2025-00149 — WIN-10058 generator slip-ring service
- WO-2025-00153 — SUB-10629 BMS firmware refresh
- WO-2025-00158 — WIN-10085 oil-spectrometer sample
- WO-2025-00161 — WIN-10016 GE firmware patch
- WO-2025-00164 — WIN-10131 high-speed shaft alignment

Most of these are due. WO-2025-00146 (cooling-loop coolant on WIN-10131) has been on the calendar since November.

Section 2 — Significant Events

1/14, 22:08 PT — WIN-10131 overheating event (first.)

Gearbox bearing temperature alarm tripped at 22:08 during a cold-weather start cycle. Unit auto-tripped. Operations brought it back online at 04:40 after a 6.5-hour outage. No emergency vendor required — crew was on site for an inspection round. See OIR-2025-005 for the full write-up.

What concerns me is that the temperature trend on WIN-10131 has been creeping for six weeks. The cooling-loop coolant exchange (WO-2025-00146) is the work order that would have caught it. That WO is on the deferred list above.

1/22 — First Apex dispatch of the year, WIN-10131.

Repeat alarm on WIN-10131 on 1/21 evening tripped the 4-hour mobilization clock under EPA-2024-11 because CPS could not commit a crew before Friday. Apex mobilized from Spokane, on site by 06:00 on 1/22. Replaced the gearbox oil cooler. Invoice \$312,000. That's about \$216K more than CPS would have charged for the same scope on a scheduled basis. I am putting it on the record now: we paid premium emergency rates for work we had on the schedule to do for \$96K.

1/26 — Inspection round, WIND-014.

GE turbines WIN-10058 and WIN-10016 both flagged elevated vibration trend on baseline comparison. Within spec per the deployed Standard MS-2025-03, but the older GE OEM manual (which is what we have to file against per the warranty) would have called for service. Inspection note logged; PM WO drafted but cannot be dispatched under the freeze.

Section 3 — Vendor Activity

- **CPS:** On site twice for scheduled WOs that pre-dated the freeze. Crew lead Maria Tonkin reports they are reducing their NW-PowerPool capacity allocation starting February if we do not put work back on their schedule. I cannot blame her — they are sized for our MSA volume and we are not feeding the MSA.
 - **Apex:** Single dispatch (1/22), WIN-10131, \$312,000. Crew was professional, fast, and they replaced the cooler. They did not refurbish it. They never do.
 - **Siemens Gamesa OEM service:** One warranty inspection on WIN-10011, no findings flagged. Their inspector noted in passing that vibration thresholds on the WIND-024 cohort "would warrant component replacement under Manual Rev. 4.2." I declined to log a service request pending guidance on MOC-2025-022.
 - **No activity:** Vestas, Nordex, Pinnacle, Heartland, ABB, Siemens Energy.
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Section 4 — Open Issues / Backlog

- 14 deferred PM WOs (listed above). Every week the freeze continues, this list will not shrink.
 - CAR-2024-118 ("Preventive Maintenance Cadence Inadequate For Aging Wind Assets") is still open from December 2024. The freeze is making it worse, not better.
 - MOC-2025-022 has not been opened yet but our engineering team is drafting it. Without that override, every Siemens Gamesa OEM inspection at WIND-024 will result in a replacement recommendation we cannot afford.
 - WIN-10131 cooling-loop work needs to happen. If the freeze runs into March I will need to either request an EPA exception or accept a higher probability of repeat overheating events.
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Section 5 — Resource & Safety Notes

Crew of 11 (6 mechanical, 3 electrical, 2 instrumentation). All current on FMP-2 fatigue tracking; no overtime exceedances in January. Weather was cold but workable — three days of high wind (>22 m/s) prevented climbing on 1/12, 1/13, and 1/27.

Two safety observations logged, neither rising to recordable. Annual respirator fit-test is overdue for two technicians; scheduled for 2/4.

Section 6 — Forward Look

If the freeze lifts February 1 as planned, we can absorb the 14 deferred WOs across February and March with one weekend crew assignment. If the freeze extends, my forecast is:

- Two to four additional first-time overheating events on WIND-024 turbines.
- One to two Apex dispatches at \$300K–\$500K each.
- Continued degradation of vibration baseline on WIND-014 GE units.

I would rather authorize \$300K of scheduled PM at CPS rates than pay \$400K to Apex for the same scope as an emergency. I have said this in writing now twice. Filing this report so it is on the record a third time.

Submitted by:

Diego Mendoza-Ruiz
Lead Field Supervisor, NW-PowerPool
Cascadia Renewable Operations LLC

References:

- OIR-2025-005 — WIN-10131 first overheating event, 2025-01-14
- CAR-2024-118 — Preventive Maintenance Cadence (open)
- EPA-2024-11 — Emergency Procurement Authorization